

Project Executable Files

Date	5 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System.
Maximum Marks	3 Marks

Project Executable Files

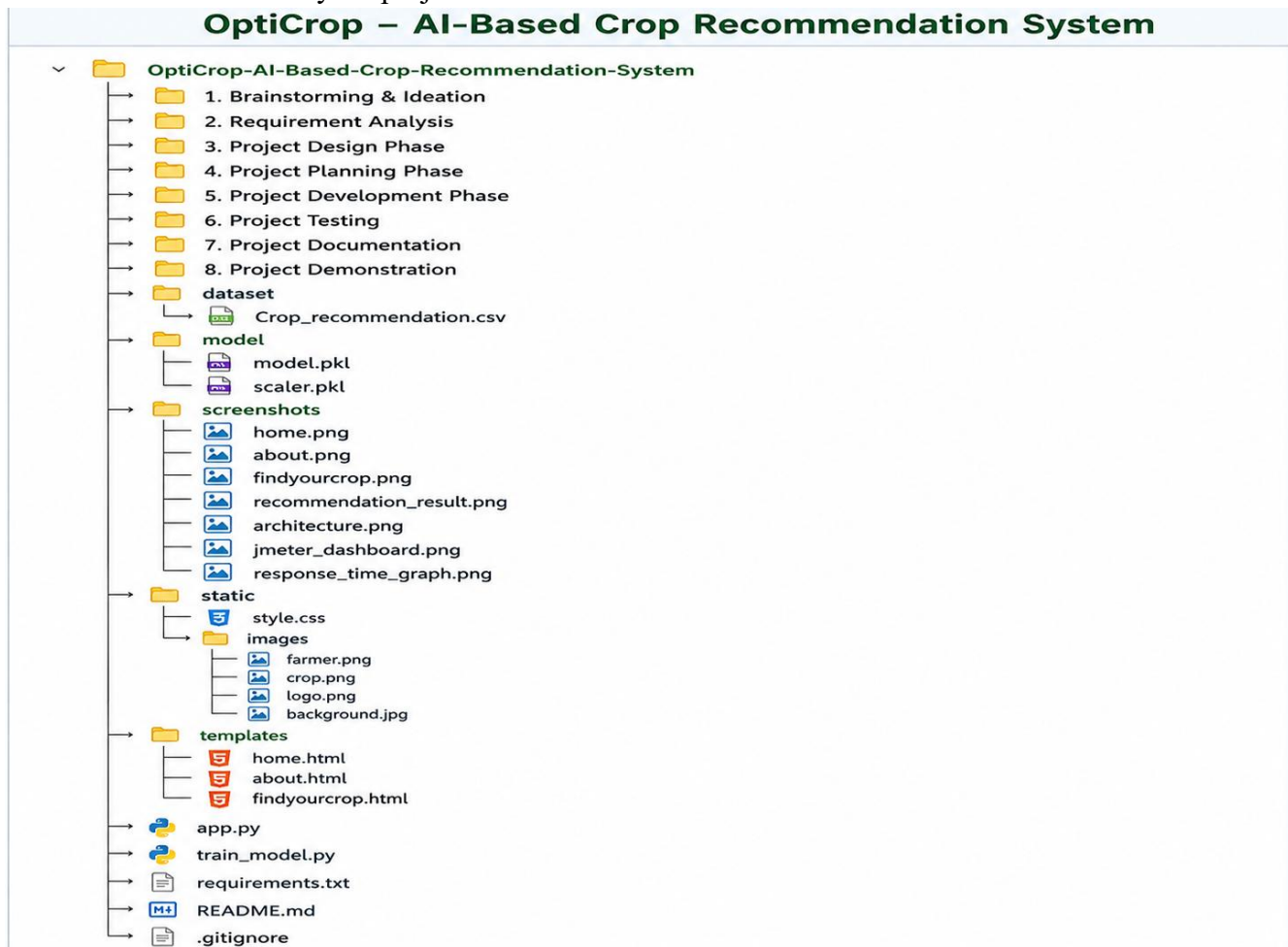
This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment or configuration file (.env.example similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:



Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opti-crop.onrender.com
Login Credentials (Demo)	Not Required
Platform /Hosting Provider	Render
Repository Link	https://github.com/Sandhali6/OPTI_CROP_AI/tree/main
Demo Video Link	

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Run Instructions – OptiCrop AI Based Crop Recommendation System 

- 1



Download or clone the OptiCrop project from the GitHub repository.

```
git clone https://github.com/23ata05317/OPTICROP.git
```
- 2



Open the project folder in Visual Studio Code or any Python IDE.

```
cd OPTICROP
```
- 3



Create a virtual environment:

```
python -m venv venv
```
- 4



Activate the virtual environment:

```
On Windows :      venv\Scripts\activate
On Mac / Linux :  source venv/bin/activate
```
- 5



Install all required dependencies:

```
pip install -r requirements.txt
```
- 6



Ensure the dataset (Crop_recommendation.csv), trained model (model.pkl), and scaler (scaler.pkl) are placed in their respective folders.
- 7



Run the Flask application:

```
python app.py
```
- 8



Open a web browser and navigate to:

```
http://127.0.0.1:5000/
```
- 9



Enter the required soil nutrient and environmental parameters and click **Predict** to view the recommended crop.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Prediction accuracy depends on the quality and size of the training dataset.	Improve the dataset and retrain the Machine Learning model.
2	The application does not include user authentication or account management.	Can be implemented in future versions.
3	On Render's free plan, the application may take a few seconds to start after being inactive.	Wait a few seconds and refresh the page if necessary.